11 A small piece of dry paper is placed inside a cylinder containing air. A tightly fitting piston is pushed rapidly into the cylinder. The paper catches fire.

Which statement best explains why the paper catches fire?

A The pressure of the air increases, causing thermal energy to be transferred from the surroundings to the paper.

B Work is done on the air, increasing its internal energy and temperature sufficiently to ignite the paper.

C The rapid compression increases the number of air molecules inside the cylinder, increasing the temperature of the air.

D The piston transfers thermal energy directly to the paper as it moves rapidly into the cylinder.